Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

The domain of the function f given by $fx = \frac{1}{\sqrt{x^2 - x - 6}}$ is

(A) $(-\infty, -2)$

(B) $(-\infty, -2) \cup [4, \infty)$

(C) $[4, \infty)$

(D) $(-\infty, -2] \cup [4, \infty)$

Q.2

The domain of definition of the function $fx = \sqrt{\log_{x-1} x^2 + 4x + 4}$, is

(A) $[-3, -1] \cup [1, 2]$

(B) $(-2, -1) \cup [2, \infty)$

(C) $(-\infty, -3] \cup (-2, -1) \cup (2, \infty)$

(D) $[-2, -1] \cup [2, \infty)$

Q.3

The domain of the function
$f(x) = \sqrt{x^{14} - x^{11} + x^6 - x^3 + x^2 + 1}$ is

(A) $(-\infty, \infty)$

(B) $[0, \infty)$

(C) $(-\infty, 0]$

(D) $\mathrm{R} \backslash [0, 1]$

Q.4

If $f(x) = \begin{cases} 2x + a & ; x \geq -1 \\ bx^2 + 3; & x < -1 \end{cases}$ and
$g(x) = \begin{cases} x + 4 & ; \quad 0 \leq x \leq 4 \\ -3x - 2; & -2 < x < 0 \end{cases}$
If domain of $g(f(x))$ is $[-1, 4]$, then $-$

(A) $\mathrm{a} = 0, \ \mathrm{b} > 5$

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(B) $\mathrm{a}=2,\ \mathrm{b}>7$

(C) $a=2, b>10$

(D) $\mathrm{a}=0,\ \mathrm{b} \in \mathrm{R}$

Q.5

The domain of the function $\sqrt{x^2-5x+6}+\sqrt{2x+8-x^2}$ is

(A) $[2,3]$

(B) $[-2,4]$

(C) $[-2,2] \cup [3,4]$

(D) $[-2,1] \cup [2,4]$

Q.6

Find the domain of
$f(x)=\sqrt{(0.625)^{4-3x}-(1.6)^{x(x+8)}}$

(A) $[-3,2]$

(B) $[1,4]$

(C) $[2,5]$

(D) $[-4,-1]$

Q.7

Find the domain of the function
$f(x)=\sqrt{\left(\frac{2}{x^2-x+1}-\frac{1}{x+1}-\frac{2x-1}{x^3+1}\right)}$

(A) $(-\infty,2]-\{-1\}$

(B) $(-\infty,2)$

(C) $]-1,2]$

(D) None of these

Q.8

The domain of $f(x)=\frac{1}{\sqrt{2x-1}}-\sqrt{1-x^2}$ is:

(A) $\left(\frac{1}{2},1\right]$

(B) $[-1,\infty[$

(C) $[1,\infty[$

(D) None of these

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Q.9

The domain of two definition of the function $f(x)$ is given by the equation $2^x + 2^y = 2$ is

(A) $0 < x \leq 1$

(B) $0 \leq x \leq 1$

(C) $-\infty < x \leq 0$

(D) $-\infty < x < 1$

Q.10

The domain of the function
$f(x) = \sqrt{x - \sqrt{1 - x^2}}$ is

(A) $\left[-1, -\frac{1}{\sqrt{2}}\right] \cup \left[\frac{1}{\sqrt{2}}, 1\right]$

(B) $[-1, 1]$

(C) $\left(-\infty, -\frac{1}{2}\right] \cup \left[\frac{1}{\sqrt{2}}, +\infty\right)$

(D) $\left[\frac{1}{\sqrt{2}}, 1\right]$

Q.11

The domain of the function $f(x) = \log_2 \left(-\log_{1/2}\left(1 + \frac{1}{x^{1/4}}\right) - 1\right)$ is

(A) $(0, 1)$

(B) $(0, 1]$

(C) $[1, \infty)$

(D) $(1, \infty)$

Q.12

The domain of the function $\mathrm{f(x)} = \frac{1}{\sqrt{{}^{10}\mathrm{C}_{\mathrm{x}-1} - 3 \times {}^{10}\mathrm{C}_{\mathrm{x}}}}$ contains the points

(A) All natural numbers

(B) Two natural numbers

(C) Two rational numbers

(D) Three rational numbers

Q.13

$f(x) = |x - 1|, f : R^+ \to R$ and $g(x) = e^x$

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$g : [-1, \infty) \to R$. If the function $fog(x)$ is defined, then its domain and range respectively are

(A) $(0, \infty)$ and $[0, \infty)$

(B) $[-1, \infty)$ and $[0, \infty)$

(C) $[-1, \infty)$ and $\left[1 - \frac{1}{e}, \infty\right)$

(D) $[-1, \infty)$ and $\left[\frac{1}{e} - 1, \infty\right)$

Q.14

The domain of the function $f(x) = {}^{24-x}C_{3x-1} + {}^{40-6x}C_{8x-10}$ is

(A) $\{2, 3\}$

(B) $\{1, 2, 3\}$

(C) $\{1, 2, 3, 4\}$

(D) None of these

Q.15

The total number of solutions of $[x]^2 = x + 2\{x\}$, where $[.]$ and $\{.\}$ denote the greatest integer function and fractional part, respectively, is equal to

(A) 2

(B) 4

(C) 6

(D) None of these

Q.16

Domain of definition of the function $f(x) = \frac{3}{4-x^2} + \log_{10}\left(x^3 - x\right)$, is

(A) $(-1, 0) \cup (1, 2) \cup (2, \infty)$

(B) $(0, 2)$

(C) $(-1, 0) \cup (0, 2)$

(D) $(1, 2) \cup (2, \infty)$

Q.17

The domain for which the functions $f(x) = 2x^2 - 1$ and $g(x) = 1 - 3x$ is equal, i.e. $f(x) = g(x)$, is

(A) $\{0, 2\}$

(B) $\left\{\frac{1}{2}, -2\right\}$

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(C) $\left\{-\frac{1}{2}, 2\right\}$

(D) $\left\{\frac{1}{2}, 2\right\}$

Q.18

The domain of the function
$f(x)=\log_e\left\{\operatorname{sgn}\left(9-x^2\right)\right\}+\sqrt{[x]^3-4[x]}$ (where
[.] represents the greatest integer function) is

(A) $[-2,1) \cup [2,3)$

(B) $[-4,1) \cup [2,3)$

(C) $[4,1) \cup [2,3)$

(D) $[2,1) \cup [2,3)$

Q.19

The domain of the real valued function $f(x)=\sqrt{5-4x-x^2}+x^2\log(x+4)$ is

(A) $(-5,1)$

(B) $-5 \leq x$ and $x \geq 1$

(C) $(-4,1]$

(D) ϕ

Q.20

The domain and range of the function f given by $\mathrm{f}(\mathrm{x})=2-|\mathrm{x}-5|$ is

(A) Domain $=\mathrm{R}^{+}$, Range $=(-\infty, 1]$

(B) Domain $=\mathrm{R}$, Range $=(-\infty, 2]$

(C) Domain $=\mathrm{R}$, Range $=(-\infty, 2)$

(D) Domain $=\mathrm{R}^{+}$, Range $=(-\infty, 2]$

Answers & Solutions

Q.1 Answer:

$-\infty, -2 \cup 4, \infty$

Solution:

Given that $fx = \frac{1}{\sqrt{x^2 - x - 6 > 0}}$ **,**

f **is defined if** $x^2 - x - 6 > 0$ **or** $x - 3x + 2 > 0$,

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$\Rightarrow x < -2$ **or** $x > 3$

$\Rightarrow x < -2$ **or** $x \geq 4$

Hence, domain $= -\infty, -2 \cup 4, \infty$

Q.2 Answer:

$-\infty, -3 \cup -2, -1 \cup 2, \infty$

Solution:

$\log_{x-1} x^2 + 4x + 4 \geq 0$

Case 1: $0 < x - 1 < 1$

i.e., $1 < x < 2$ **, then**

$0 < x^2 + 4x + 4 \leq 1 \Rightarrow x^2 + 4x + 3 \leq 0 \ \& \ x + 2^2 > 0$

$\Rightarrow -3 \leq x \leq -1 \ \& \ x \neq -2$

So, $x \in -2, -1$

Case 2: $x - 1 > 1$

i.e., $x > 2$

$x^2 + 4x + 4 \geq 1 \Rightarrow x + 1x + 3 \geq 0 \Rightarrow x \in -\infty, -3 \cup -1, \infty$

$\Rightarrow x \in -\infty, -3 \cup 2, \infty$

Hence, domain is $-\infty, -3 \cup (-2, -1) \cup 2, \infty$

Q.3 Answer:

$(-\infty, \infty)$

Solution:

Given $f(x) = \sqrt{x^{14} - x^{11} + x^6 - x^3 + x^2 + 1}$

for $f(x)$ **to be defined,**

$x^{14} - x^{11} + x^6 - x^3 + x^2 + 1 \geq 0$

Case $1 : x \geq 1$

$x^{14} - x^{11} + x^6 - x^3 + x^2 + 1$

$= (x^{14} - x^{11}) + (x^6 - x^3) + (x^2 + 1) > 0$

Case 2: $0 \leq x \leq 1$

$x^{14} - x^{11} + x^6 - x^3 + x^2 + 1$

$= x^{14} \{(x^{11} - x^{11}) + (x^3 - x^2) + 1 > 0$

$\{\because x^{11} - x^6 \leq 0, x^3 - x^2 \leq 0\}$

Case $3 : x < 0 \ x^{14} - x^{11} + x^6 - x^3 + x^2 + 1 > 0$

$(\because x^{11} < 0, x^3 < 0, x^{14}, x^6, x^2 > 0)$

Thus for all area, $x, x^{14} - x^{11} + x^6 - x^3 + x^2 + 1 \geq 0$

Hence the domain of $f(x) = \mathrm{R} = (-\infty, \infty)$

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Q.4 Answer:

$a = 0,\ b > 5$

Solution:

$f(4) = g(4) \Rightarrow 8 + a = 8 \Rightarrow a = 0$
$f(-1) = -2$ **for** $a = 0$
$f(-1) > f(4)$
$b + 3 > 8 \Rightarrow b > 5$

Q.5 Answer:

$[-2, 2] \cup [3, 4]$

Solution:

$f(x) = \sqrt{(x-2)(x-3)} + \sqrt{-(x-4)(x+2)}$
The first part is real outside $(2, 3)$ **and the second is real in** $[-2, 4]$ **so that the domain is** $[-2, 2] \cup [3, 4]$.

Q.6 Answer:

$[-4, -1]$

Solution:

Clearly, $(0.625)^{4-3x} \geq (1.6)^{x(x+8)}$
or $\left(\frac{5}{8}\right)^{4-3x} \geq \left(\frac{8}{5}\right)^{x(x+8)}$
or $\left(\frac{8}{5}\right)^{3x-4} \geq \left(\frac{8}{5}\right)^{x(x+8)}$
or $3x - 4 \geq x^2 + 8x$ **or** $x^2 + 5x + 4 \leq 0$ **or** $-4 \leq x \leq -1$
Hence, the domain of function $f(x)$ **is** $x \in [-4, -1]$

Q.7 Answer:

$(-\infty, 2] - \{-1\}$

Solution:

$f(x) = \sqrt{\left(\frac{2}{x^2-x+1} - \frac{1}{x+1} - \frac{2x-1}{x^3+1}\right)}$
We must have $\frac{2}{x^2-x+1} - \frac{1}{x+1} - \frac{2x-1}{x^3+1} \geq 0$
or $\frac{2(x+1)-\left(x^2-x+1\right)-(2x-1)}{(x+1)(x^2-x+1)} \geq 0$
or $\frac{-\left(x^2-x-2\right)}{(x+1)(x^2-x+1)} \geq 0$
or $\frac{-(x-2)(x+1)}{(x+1)(x^2-x+1)} \geq 0$
or $\frac{2-x}{x^2-x+1} \geq 0$, **where** $x \neq -1$
or $\quad 2 - x \geq 0, x \neq -1$ (**as** $x^2 - x + 1 > 0 \forall x \in R$)
or $\quad x \leq 2, x \neq -1$
Hence, domain of the function is $(-\infty, -1) \cup (-1, 2]$ **or** $(-\infty, 2] - \{-1\}$

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Q.8 Answer:

$\left(\frac{1}{2}, 1\right]$

Solution:

Given, $f(x)=\frac{1}{\sqrt{2x-1}}-\sqrt{1-x^2}=p(x)-q(x)$
where $p(x)=\frac{1}{\sqrt{2x-1}}$ and $q(x)=\sqrt{1-x^2}$
Now, Domain of $p(x)$ exist when $2x-1 \neq 0$
$\Rightarrow \quad x=\frac{1}{2}$ and $2x-1>0$
$\Rightarrow \quad x=\frac{1}{2}$ and $x>\frac{1}{2} \therefore x \in \left(\frac{1}{2}, \infty\right)$
and domain of $q(x)$ exists when $1-x^2 \geq 0$
$\Rightarrow x^2 \leq 1 \Rightarrow |x| \leq 1 \quad \therefore -1 \leq x \leq 1$
$\therefore \quad$ Common domain is $\left(\frac{1}{2}, 1\right]$

Q.9 Answer:

$-\infty < x < 1$

Solution:

We have, $2^x+2^y=2 \Rightarrow 2^y=2-2^x$
$\Rightarrow y=\log_2(2-2^x) \Rightarrow 2-2^x>0 (\therefore y \text{ is real})$
$\Rightarrow 2^x<2 \Rightarrow 2^{x-1}<1 \Rightarrow x-1<0 \Rightarrow x<1$
$\Rightarrow -\infty < x < 1$

Q.10 Answer:

$\left[\frac{1}{\sqrt{2}}, 1\right]$

Solution:

For $f(x)$ to be defined, we must have $x-\sqrt{1-x^2} \geq 0$ or $x \geq \sqrt{1-x^2} > 0$ or $x^2 \geq 1-x^2$ or $x^2 \geq \frac{1}{2}$.
Also, $1-x^2 \geq 0$ or $x^2 \leq 1$
Now, $x^2 \geq \frac{1}{2} \Rightarrow \left(x-\frac{1}{\sqrt{2}}\right)\left(x+\frac{1}{\sqrt{2}}\right) \geq 0$
$\Rightarrow x \leq -\frac{1}{\sqrt{2}}$ or $\quad x \geq \frac{1}{\sqrt{2}}$
Also, $x^2 \leq 1 \Rightarrow (x-1)(x+1) \leq 0 \Rightarrow -1 \leq x \leq 1$
Thus, $x>0, x^2 \geq \frac{1}{2}$ and $x^2 \leq 1 \Rightarrow x \in \left[\frac{1}{\sqrt{2}}, 1\right]$

Q.11 Answer:

$(0, 1)$

Solution:

$f(x)$ is defined if $-\log_{1/2}\left(1+\frac{1}{x^{1/4}}\right)-1>0$
$\Rightarrow \log_{1/2}\left(1+\frac{1}{x^{1/4}}\right)<-1$

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$\Rightarrow \quad 1+\frac{1}{x^{1/4}}>\left(\frac{1}{2}\right)^{-1}$

$\Rightarrow \quad \frac{1}{x^{1/4}}>1 \Rightarrow 0<x<1$

Q.12 Answer:

Two natural numbers

Solution:

The expression inside the square root must not be negative.

change

The condition is ${}^{10}C_{x-1}>3{}^{10}C_{x}$

This is true for $x=10$ **and for** $x=9$ **only.**

It could not have been true till $x=5$ **as the number only increases till it reaches** $\frac{n}{2}$

Q.13 Answer:

$[-1,\infty)$ and $[0,\infty)$

Solution:

$$f(x)=|x-1|=\begin{cases}1-x, 0<x<1\\ x-1, x\geq 1\end{cases}$$

$g(x)=e^{x}, x\geq -1$

$$(\text{ fog })(x)=\begin{cases}1-g(x), & 0<g(x)<1 \text{ i.e. } -1\leq x<0\\ g(x)-1, & g(x)\geq 1 \text{ i.e. } 0\leq x\end{cases}$$

$$=\begin{cases}1-e^{x}, & -1\leq x<0\\ e^{x}-1, & x\geq 0\end{cases}$$

$\therefore$ **domain** $=[-1,\infty)$

fog is decreasing in $[-1,0)$ **and increasing in** $[0,\infty)$

$\text{fog}(-1)=1-\frac{1}{e}$ **and** $\text{fog}(0)=0$

As $x\to\infty, \text{fog}(x)\to\infty$

$\therefore$ **range** $=[0,\infty)$

Q.14 Answer:

$\{2,3\}$

Solution:

${}^{24-x}C_{3x-1}$ **is defined if,** $24-x>0, 3x-1\geq 0$ **and** $24-x\geq 3x-1$

$\Rightarrow x<24, x\geq\frac{1}{3}$ **and** $x\leq\frac{25}{4} \Rightarrow \frac{1}{3}\leq x\leq\frac{25}{4}$

${}^{40-6x}C_{8x-10}$ **is defined if**

$40-6x>0, 8x-10\geq 0$ **and** $40-6x\geq 8x-10$

$\Rightarrow \quad x<\frac{20}{3}, x\geq\frac{5}{4}$ **and** $x\leq\frac{25}{7}$

$\Rightarrow \frac{5}{4}\leq x\leq\frac{25}{7}$

(1) and (2), we get $\frac{5}{4}\leq x\leq\frac{25}{7}$ **From**

But $24-x\in N, \therefore x$ **must be an integer,** $x=2,3$ **Hence domain** $(f)=\{2,3\}$

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Q.15 Answer:

4

Solution:

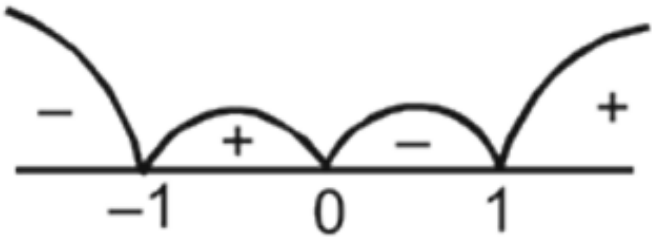
$[x]^2 = x + 2\{x\}$
$\Rightarrow [x]^2 = [x] + 3\{x\}$
$\Rightarrow \{x\} = \frac{[x]^2 - [x]}{3}$
$\Rightarrow 0 \leq \frac{[x]^2 - [x]}{3} < 1$
$\Rightarrow 0 \leq [x]^2 - [x] < 3$
$\Rightarrow [x] \in \left(\frac{1-\sqrt{13}}{2}, 0\right] \cup \left[1\frac{1+\sqrt{13}}{2}\right)$
$\Rightarrow [x] = -1, 0, 1, 2$
$\Rightarrow \{x\} = \frac{2}{3}, 0, 0, \frac{2}{3}$, **(respectively)**
$\Rightarrow x = -\frac{1}{3}, 0, 1, \frac{8}{3}$

Q.16 Answer:

$(-1, 0) \cup (1, 2) \cup (2, \infty)$

Solution:

$f(x) = \frac{3}{4-x^2} + \log_{10}(x^3 - x)$
$4 - x^2 \neq 0; x^3 - x > 0$
$x \neq \pm\sqrt{4}$ **and** $x(x+1)(x-1)$
$\therefore D = (-1, 0) \cup (1, \infty) - \{\sqrt{4}\}$
$D = (-1, 0) \cup (1, 2) \cup (2, \infty).$



Q.17 Answer:

$\left\{\frac{1}{2}, -2\right\}$

Solution:

For $f(x) = g(x) \Rightarrow 2x^2 - 1 = 1 - 3x$
$\Rightarrow 2x^2 + 3x - 2 = 0 \Rightarrow 2x^2 + 4x - x - 2 = 0$
$\Rightarrow \quad 2x(x+2) - 1(x+2) = 0 \Rightarrow (x+2)(2x-1) = 0$
$\Rightarrow \quad x = -2, \frac{1}{2}$
$\therefore$ **The domain for which the function**
$f(x) = g(x)$ **is** $\left\{-2, \frac{1}{2}\right\}$

Q.18 Answer:

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$[-2,1) \cup [2,3)$

Solution:

We have $f(x)=\log_e\{\operatorname{sgn}(9-x^2)\}+\sqrt{[x]^3-4[x]}$

We must have, $\operatorname{sgn}(9-x^2)>0 \Rightarrow 9-x^2>0$

$\Rightarrow x^2-9<0 \Rightarrow (x-3)(x+3)<0 \Rightarrow -3<x<3$

Also $[x]^3-4[x] \geq 0 \Rightarrow [x]\left([x]^2-4\right) \geq 0$

$\Rightarrow [x]([x]-2)([x]+2) \geq 0$

$\Rightarrow [x] \geq -2$ **or** $[x]$ **lies between** -2 **and 0**

i.e., $[x]=-2,-1$ **or 0 Now,** $[x] \geq -2 \Rightarrow x \geq 2$

(ii) $[x]=-2 \Rightarrow -2 \leq x<-1; [x]=-1 \Rightarrow -1 \leq x<0$

$[x]=0 \Rightarrow 0 \leq x<1$ **Hence** $[x]=-2,-1,0$

$\Rightarrow -2 \leq x<1$

Hence $D_f=[-2,1) \cup [2,3)$.

Q.19 Answer:

$(-4,1]$

Solution:

$f(x)=\sqrt{5-4x-x^2}+x^2\log(x+4)$

$\Rightarrow 5-4x-x^2 \geq 0, x+4>0$

$\Rightarrow \quad (x+5)(x-1) \leq 0, x>-4$

$\Rightarrow \quad -5 \leq x \leq 1, x>-4 \Rightarrow -4<x \leq 1$

Q.20 Answer:

Domain $=\mathrm{R}$, Range $=(-\infty,2]$

Solution:

Given $\mathrm{f(x)}=2-|\,\mathrm{x}-5\,|$

Domain of $\mathrm{f(x)}$ **is defined for all real values of** x **Since,** $|\mathrm{x}-5| \geq 0 \Rightarrow -|\mathrm{x}-5| \leq 0$

$\Rightarrow 2-|x-5| \leq 2 \Rightarrow f(x) \leq 2$

Hence, range of $\mathrm{f(x)}$ **is** $(-\infty,2]$

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